Which artifacts does Déjà View show?

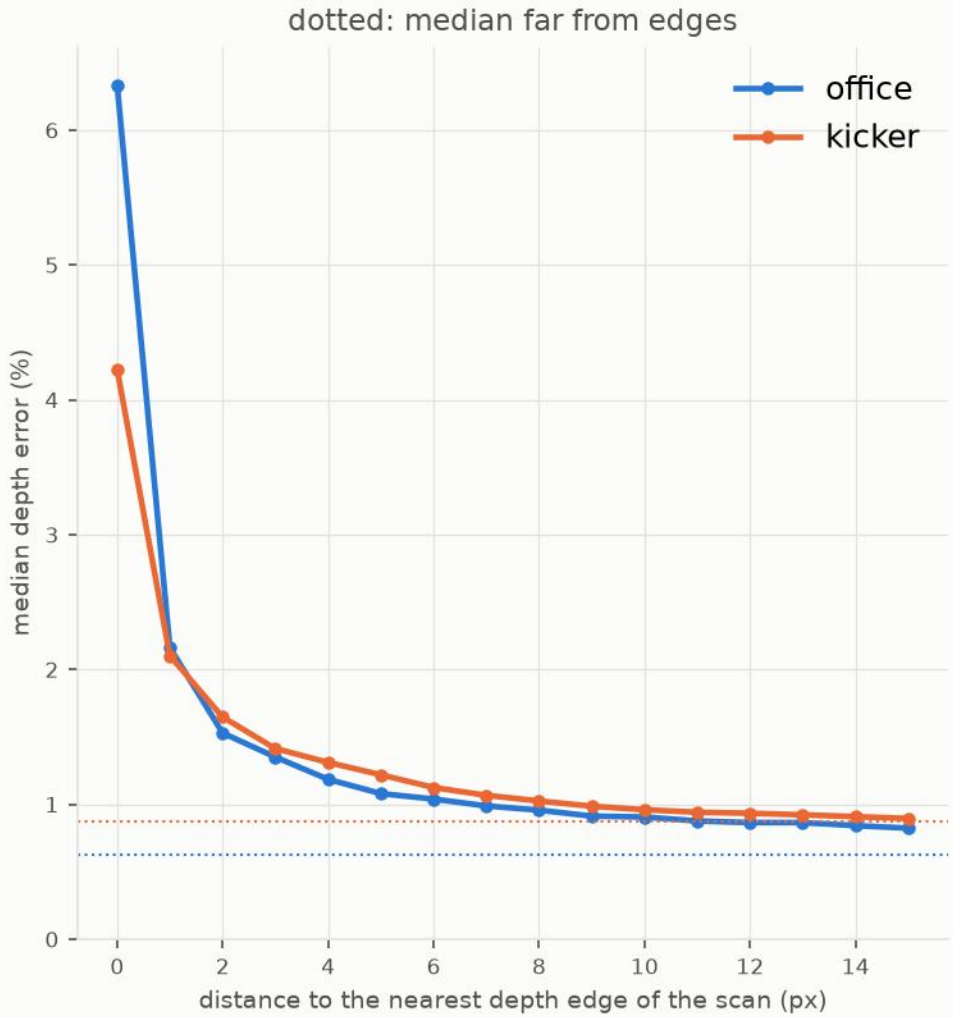
Office / kicker, all 20 views each, against ETH3D's laser scans; each depth map scaled by its median ratio to the scan.

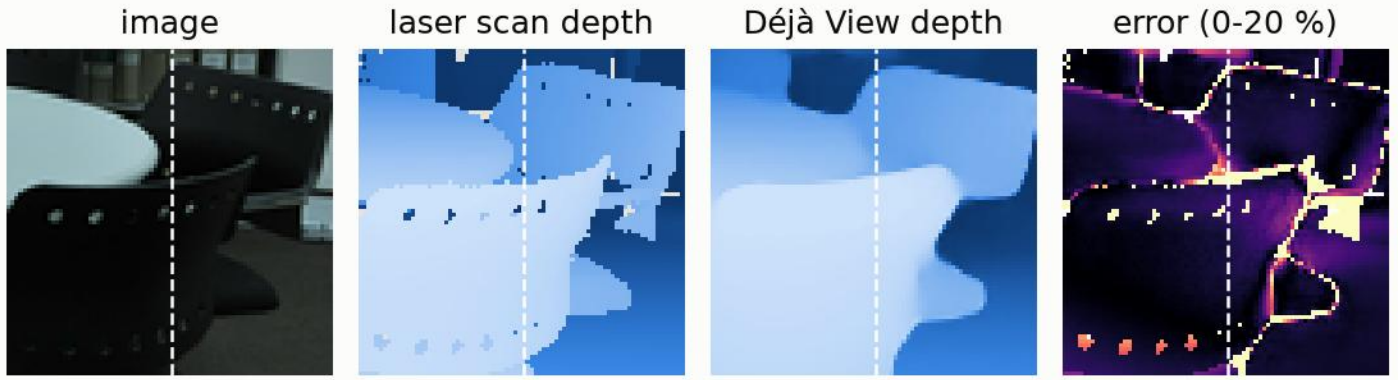
Depth bleeding, flying pixels	**confirmed**	median error 6.3 / 4.2 % at depth edges vs 0.6 / 0.9 % away
Wavy or thick planes	**confirmed**	0.6 cm RMS from a plane, scan 0.11 cm; 14 px patch grid in the ray map (score 1.8 / 1.9)
Cross-view drift, layering	**confirmed**	views disagree by 0.4 / 0.8 %; per-view offsets spread 3.3 / 2.8 %
Textureless regions	**not confirmed**	flattest vs most textured tenth: 0.5 vs 0.8 % (office), 1.1 vs 0.9 % (kicker)
Reflective, transparent	**not measurable**	no laser return on 17 / 12 % of pixels (windows, lamps, dark surfaces)
Confidence misses errors	**partly**	Spearman -0.47 / -0.27; the confident half keeps 45 / 66 % of the error

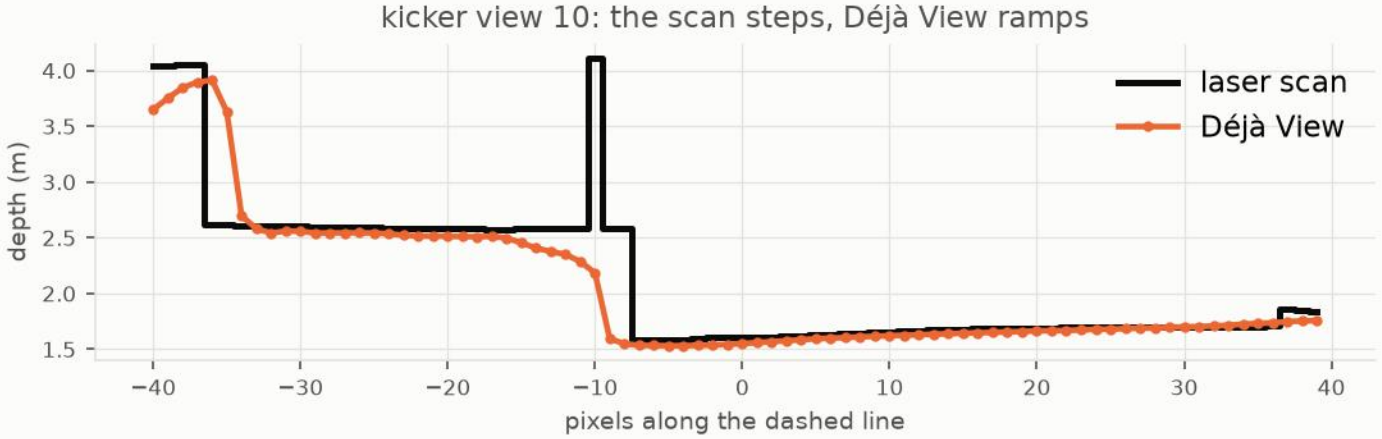
Déjà View (K = 16) on 20 views each of ETH3D office and kicker; ground truth: ETH3D laser scans (CC BY-NC-SA 4.0).

1 · Depth bleeding at object boundaries: confirmed

Median depth error 6.3 % (office) and 4.2 % (kicker) on pixels at a depth edge, 1.1 / 1.2 % five pixels away, 0.6 / 0.9 % far from edges. 54 / 47 % of edge pixels are off by more than 5 %: they float between foreground and background.





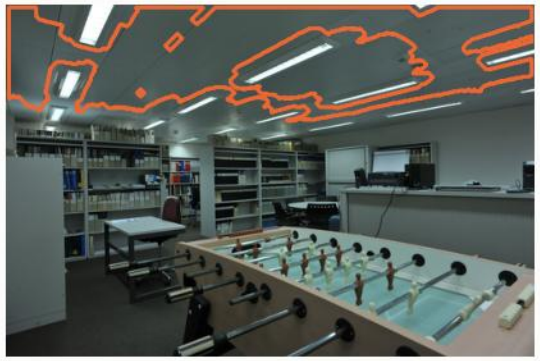


Déjà View (K = 16) on 20 views each of ETH3D office and kicker; ground truth: ETH3D laser scans (CC BY-NC-SA 4.0). Depth edge: the scan's log depth jumps by more than 5 % between neighbors.

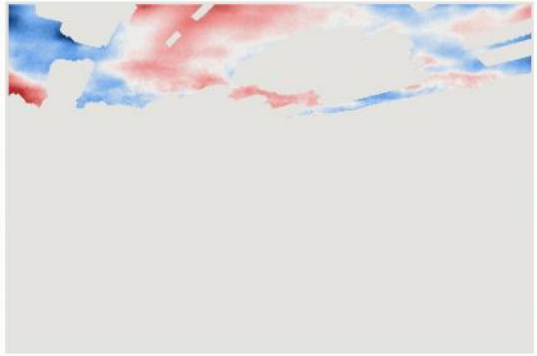
2 · Wavy planes: confirmed

On walls, floors, ceilings and tables, Déjà View's points lie 0.6 cm RMS from their best-fit plane (median over 105 view-planes; worst tenth over 3 cm); the scan's 0.11 cm. Its 14 px patch grid is in the ray map, not the depth: grid score 1.8 / 1.9 along its own rays, 0.9 / 0.9 along a pinhole (1 = no grid).

kicker view 9: one plane (outlined)



Déjà View: RMS 0.6 cm

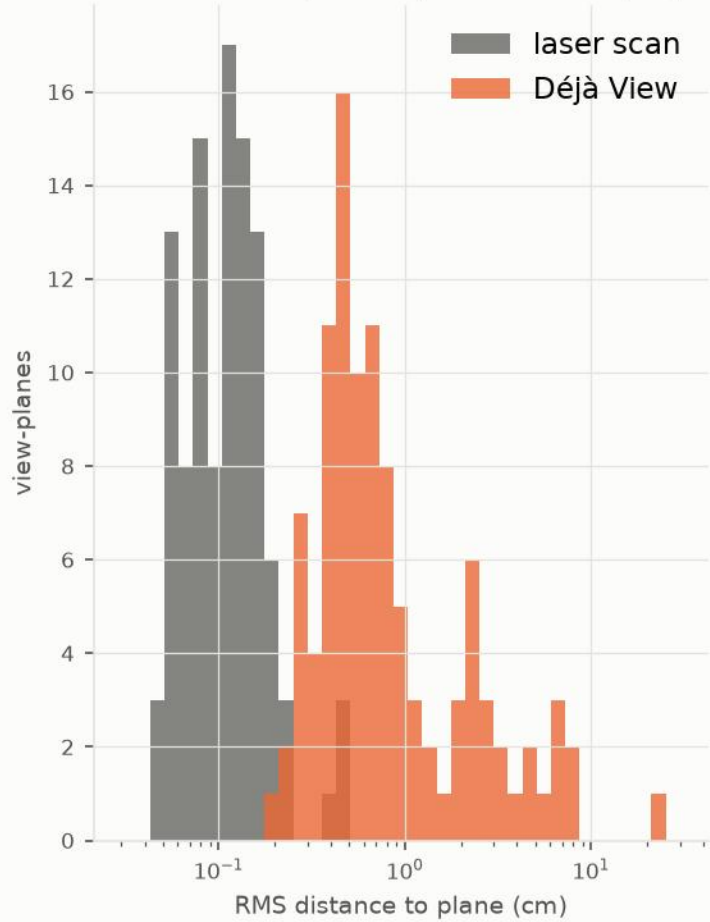


laser scan: RMS 0.18 cm





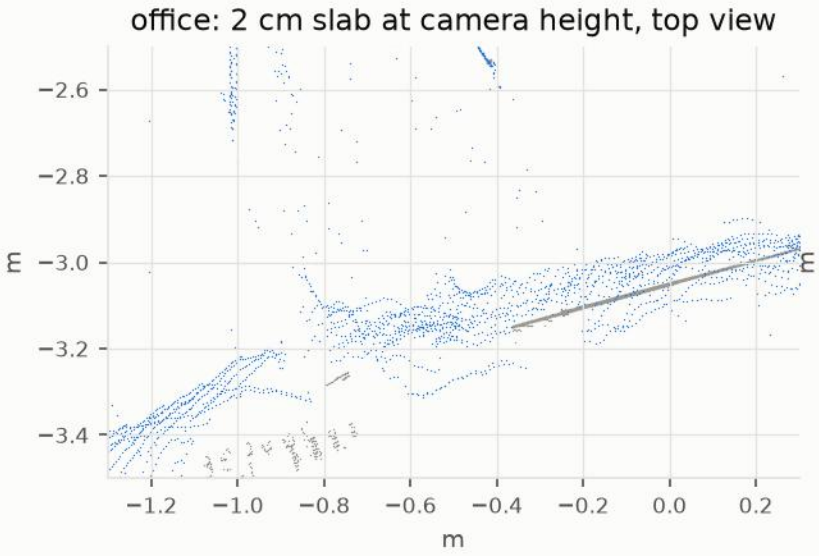
all 105 view-planes (median example)

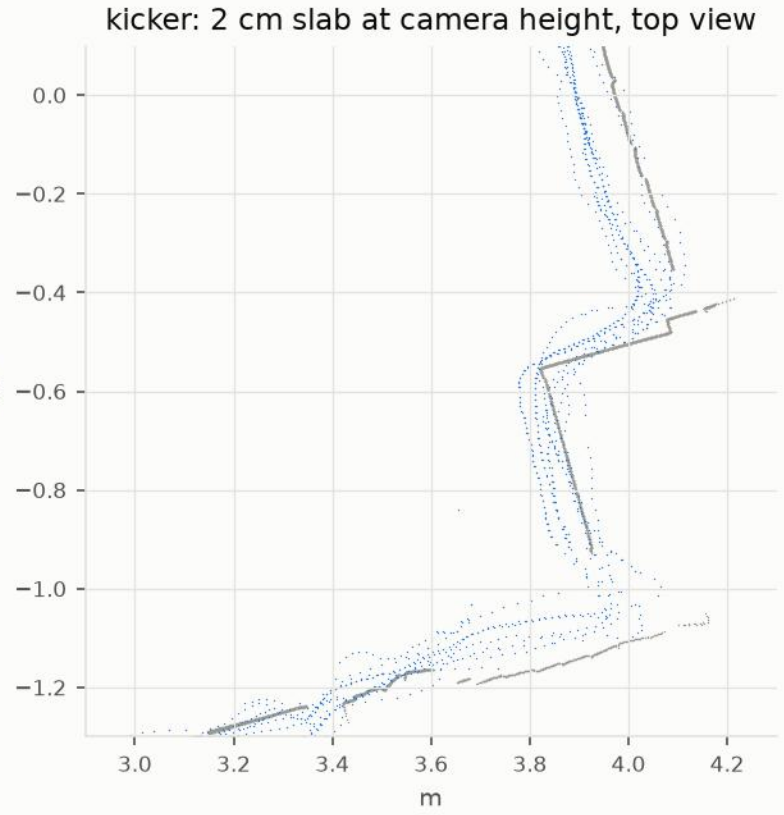


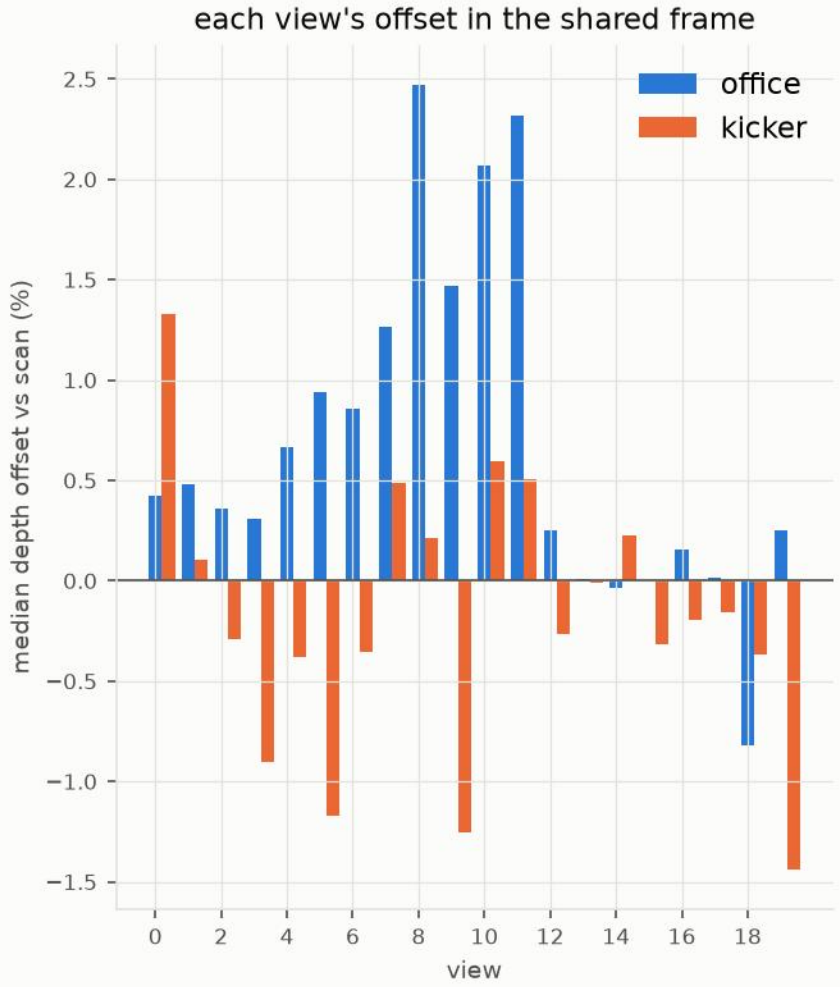
Déjà View (K = 16) on 20 views each of ETH3D office and kicker; ground truth: ETH3D laser scans (CC BY-NC-SA 4.0). Planes: RANSAC on the scan (1 cm); per view, the largest connected region within 1 cm, eroded by 3 px.

3 · Cross-view drift layers the surfaces: confirmed

Overlapping views disagree by 0.4 % (office) and 0.8 % (kicker) of depth, median; each view sits at its own offset (spread 3.3 / 2.8 %). Fused, every wall comes out several times. Gray: laser scan; blue: all 20 Déjà View depth maps.



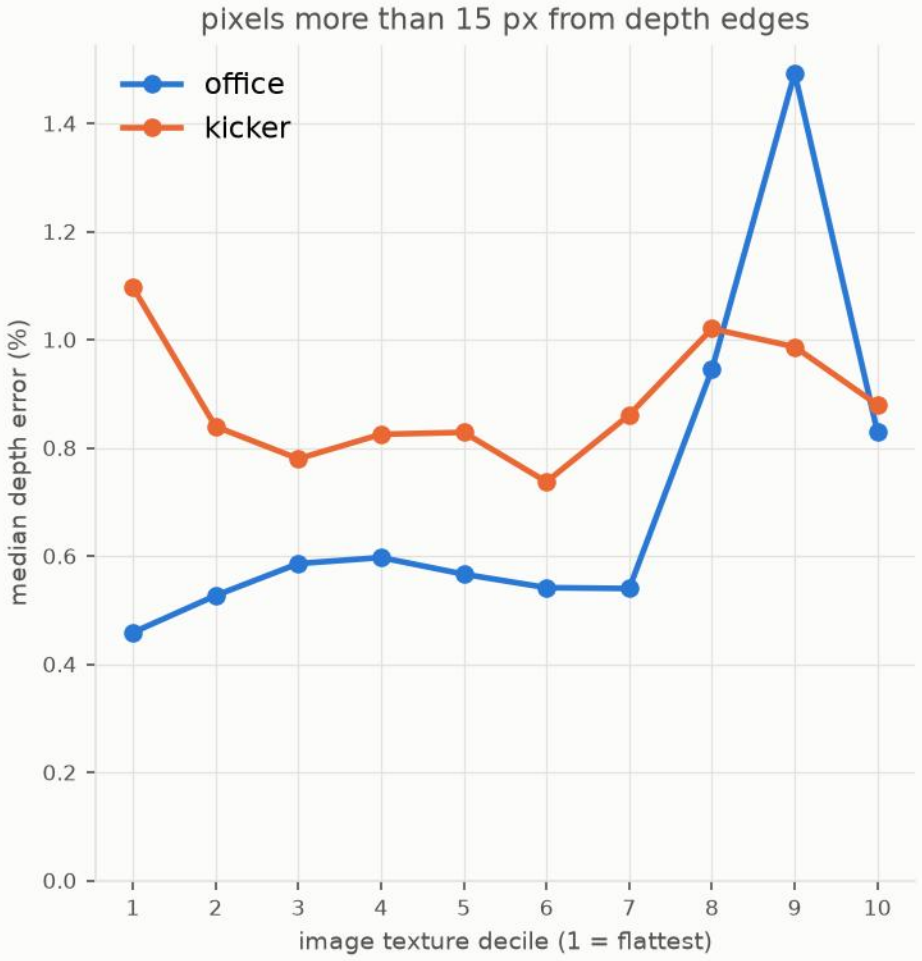




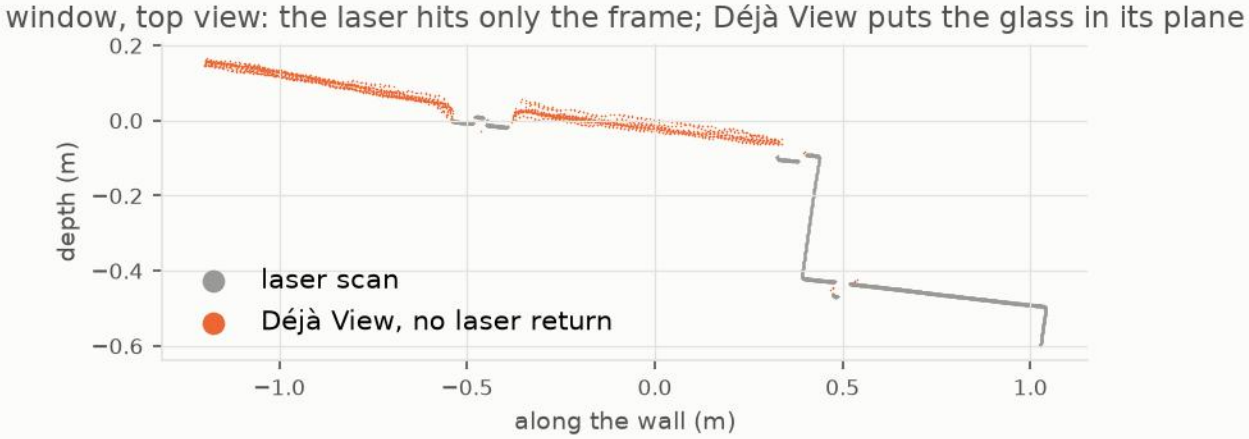
Déjà View (K = 16) on 20 views each of ETH3D office and kicker; ground truth: ETH3D laser scans (CC BY-NC-SA 4.0). Fused cloud mapped to the scan by one similarity on the camera centers.

4 · Textureless: not confirmed · reflective: not measurable

Away from depth edges, textureless regions are no worse than textured ones (office: 0.5 % in the flattest tenth vs 0.8 % in the most textured; kicker 1.1 vs 0.9 %). Glass and lamps return no laser, so ground truth is missing exactly there.



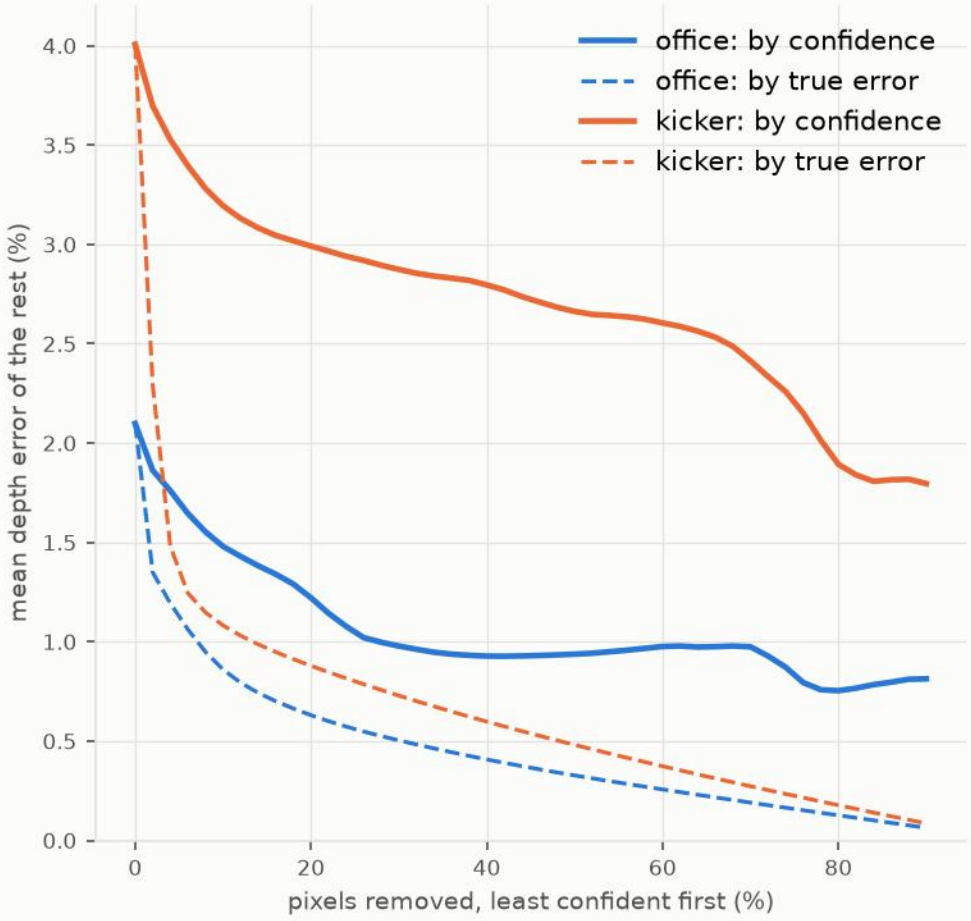




Déjà View (K = 16) on 20 views each of ETH3D office and kicker; ground truth: ETH3D laser scans (CC BY-NC-SA 4.0). Texture: gray-level std in 7 x 7 px, deciles per scene.

5 · Confidence locates the errors only partly

Rank correlation of confidence with error: -0.47 (office), -0.27 (kicker). Keeping the most confident half removes 55 / 34 % of the mean error; knowing the true error, the best half would remove 84 / 88 %.



kicker view 10

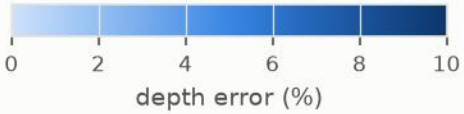


confidence (dark = high)



true error (gray: no scan)





Déjà View (K = 16) on 20 views each of ETH3D office and kicker; ground truth: ETH3D laser scans (CC BY-NC-SA 4.0).